

MD. ISTIAK SOBHAN, PhD

Environmental & Social Safeguards | Climate Finance | ESG and Responsible Finance Research

Dhaka, Bangladesh | Nationality: Bangladeshi | Date of Birth: December 1964

PROFILE SUMMARY

Environmental scientist and development finance specialist with over 35 years of professional experience in environmental and social due diligence, safeguards policy application, natural resource management and climate-related policy research in Bangladesh and across Asia, the Middle East and southern Africa.

Served for approximately 12 years with the World Bank in Bangladesh, most recently as Senior Environmental Specialist, applying the World Bank Environmental and Social Framework (ESF) and predecessor Safeguard Policies across a portfolio of more than 20 investment operations, and acting as Task Team Leader for large national programmes in forestry, coastal and marine fisheries. Holds World Bank accreditation in Safeguards, ESF, Investment Project Financing TTL and Program-for-Results TTL.

Deep familiarity with the design, appraisal, supervision and grievance/accountability arrangements of MDB-financed mega projects, combined with a strong record of peer-reviewed research on climate change impacts on vulnerable coastal communities, salinisation, and ecosystem-dependent livelihoods in the Sundarbans and the wider Bangladesh delta. Extensive experience of qualitative and quantitative field research, key informant interviews, community-level evidence synthesis, and production of policy reports and advocacy-ready briefs for government, civil society and donor audiences.

RELEVANCE TO THIS ASSIGNMENT

- **MDB safeguard frameworks:** Accredited practitioner of the World Bank ESF and the earlier Operational Policies; working knowledge of the ADB Safeguard Policy Statement, AIIB Environmental and Social Framework and IFC Performance Standards through co-financed operations.
- **Mega project experience:** Direct engagement with large-scale MDB-financed infrastructure, energy, water, forestry and fisheries operations in Bangladesh, including screening, ESIA/ESMP review, category determination, supervision and compliance monitoring across the full project lifecycle.
- **Climate justice lens:** Published research on differentiated impacts of climate change, salinity intrusion and sea-level rise on poor and coastal communities, fish habitats and mangrove-dependent livelihoods, with attention to gendered and marginalised-group outcomes.
- **Accountability and grievance mechanisms:** Practical experience of project-level grievance redress systems, stakeholder engagement plans, information disclosure requirements and the interface with MDB independent accountability mechanisms.
- **Policy research and advocacy products:** Lead or contributing author of national reports, management plans and World Bank policy research working papers; experienced in

producing concise briefs for regulators and policymakers and presenting findings in validation workshops.

- **National networks:** Established working relationships with Government of Bangladesh ministries and line agencies, IUCN and civil society organisations, academia and development partners, supporting efficient access to key informants.

EDUCATION

- PhD, Natural Resources Management — Wageningen University and the International Institute for Geo-Information Science and Earth Observation (ITC), The Netherlands, 2007
- MSc, Environmental Systems Analysis and Monitoring (with distinction) — ITC, The Netherlands, 1999
- MSc, Botany (First Class) — University of Dhaka, Bangladesh, 1987
- BSc (Honours), Biology / Botany with Biochemistry and Soil Science (First Class) — University of Dhaka, Bangladesh, 1986

PROFESSIONAL ACCREDITATION AND MEMBERSHIP

- World Bank: Safeguard Accreditation; Environmental and Social Framework (ESF) Accreditation; Task Team Leader Accreditation for Investment Project Financing; Task Team Leader for Program-for-Results; Trust Fund Accreditation for TTLs (v2.0)
- Member, International Society for Photogrammetry and Remote Sensing; Member, Society of Photo-Optical Instrumentation Engineers
- Life Member, Bangladesh Botanical Society; Member, Asiatic Society of Bangladesh; Member, Bangladesh National Biodiversity Group; Member, ITC Alumni Association, The Netherlands

TECHNICAL AND RESEARCH SKILLS

- Environmental and social due diligence: ESIA/EIA and ESMP review, safeguard screening and categorisation, environmental and social audits, compliance monitoring, stakeholder engagement and grievance redress design
- Research methods: biophysical and socio-economic survey design, sampling design, structured household surveys, key informant interviews, focus group discussions, case study development
- Digital data collection: KoboToolbox, ArcGIS Survey123, GPS/GPS-Map
- Analysis: R, SPSS, Python, C#; statistics, data mining and machine learning; ESRI Geodatabase and PostGIS; remote sensing with ERDAS, ENVI and eCognition

COUNTRIES OF WORK EXPERIENCE

Bangladesh, United Arab Emirates, Bhutan, India, Nepal, Zimbabwe and The Netherlands

LANGUAGES

English — Excellent (speaking, reading, writing)

Bengali — Native (speaking, reading, writing)

Hindi — Basic (speaking, reading); Urdu — Basic (speaking)

EMPLOYMENT RECORD

August 2021 – July 2026 | Senior Environmental Specialist

World Bank, Bangladesh Country Office, Dhaka

- Task Team Leader for the Sustainable Forests and Livelihoods (SUFAL) Project and the Sustainable Coastal and Marine Fisheries Project, with responsibility for design, procurement oversight, implementation supervision, results monitoring and dialogue with Government counterparts.
- Provided environmental and social risk management support to a portfolio of more than 20 Bank-financed operations across energy, transport, water, urban, forestry and fisheries sectors, covering screening, risk classification, ESIA/ESMF review, ESCP monitoring and corrective action planning under the ESF.
- Assessed distribution of project costs, benefits and risks across affected groups, with specific attention to coastal, forest-dependent and climate-vulnerable communities, women and Indigenous Peoples.
- Reviewed and strengthened project-level grievance redress mechanisms, stakeholder engagement plans and information disclosure practice; supported borrower capacity building on safeguards and ESF implementation.
- Contributed to analytical work and policy dialogue on climate finance, coastal resilience and natural resource governance, including Bank policy research publications on salinisation and climate impacts in the Sundarbans.

August 2016 – July 2021 | Environmental Specialist

World Bank, Bangladesh Country Office, Dhaka

- Led environmental due diligence for investment operations, including review of ESIA's, environmental management plans and monitoring reports prepared by implementing agencies.
- Supported the transition of the country portfolio from Operational Policies to the Environmental and Social Framework, including gap analysis, borrower capacity assessment and preparation of Environmental and Social Commitment Plans.
- Undertook field supervision missions to project-affected areas, documenting community-level impacts and engaging with local government, affected households and civil society.

September 2014 – July 2016 | Short Term Consultant

World Bank, Bangladesh Country Office, Dhaka

- Provided technical inputs to environmental assessment, climate change and natural resource management tasks, including analytical studies on climate change impacts on ecosystems and coastal livelihoods.

July 2010 – August 2014 | Programme Coordinator

IUCN — International Union for Conservation of Nature, Bangladesh Country Office, Dhaka

- Led programme development, monitoring and evaluation for the country office, managing a team of more than 15 senior and mid-level professionals; the office held the second largest portfolio in the IUCN Asia region during this period.
- Directed proposal development, fundraising and donor engagement across climate change adaptation, species conservation, mangrove and semi-evergreen forest biodiversity management, Ramsar site management and transboundary water dialogue.
- Coordinated multi-stakeholder policy engagement with government, donors, academia and civil society coalitions.

August 2009 – June 2010 | Principal Nature Conservation Officer

Dubai Municipality, Government of the United Arab Emirates

- Oversaw biodiversity and nature conservation activities across the Emirate of Dubai, including protected area management and management of Ras Al Khor, a Ramsar site; prepared conservation plans and budgets and supervised a team of eight.

July 2007 – July 2009 | Director, Ecology Division

Center for Environmental and Geographic Information Services (CEGIS), Dhaka

- Directed the Ecology Division, supervising impact analysis of development projects on ecosystems, natural resource and ecosystem management assignments, and project monitoring for government and donor clients.

April 2003 – June 2007 | PhD Researcher

ITC / Wageningen University, The Netherlands

- Doctoral research on plant species discrimination using hyperspectral remote sensing at laboratory and landscape scale, with fieldwork in Parco della Majella, Abruzzo, Italy, including analysis of protected area management regimes in fragmented and heterogeneous landscapes.

June 2000 – March 2003 | Ecologist

Environmental and GIS Support Project for Water Sector Planning (EGIS II) / CEGIS, Dhaka

- Analysed the ecological impacts of water sector development projects and contributed to land use planning and environmental assessment for national water sector investments.

April 1999 – May 2000 | Environmentalist

Resource Planning & Management Consultants (Pvt.) Ltd. (RPMC), Dhaka — Dampara Water Management Project (CIDA-funded)

- Reviewed feasibility-stage EIA and EMP, reassessed environmental impacts of proposed works, identified new and altered environmental issues, recommended mitigation and enhancement measures, and designed baseline monitoring survey instruments.

September 1997 – March 1999 | MSc Researcher

ITC, The Netherlands

- Research on spatial distribution and environmental boundaries of miombo and mopane woodlands in Karoi and Hurungwe districts, Mashonaland West, Zimbabwe, using GIS modelling.

February 1997 – August 1997 | Wetland Project Officer

IUCN Bangladesh Country Office, Dhaka (Ford Foundation-funded)

- Supported formulation of national wetland policy and legislation for EIA activities involving wetlands; prepared consultant terms of reference, organised consultations among consultants, stakeholders and line ministries, and consolidated reporting.

December 1995 – March 1996 | Resource Planner (part time)

Canarail Consultant International Ltd., Dhaka — Jamuna Bridge Railway Approaches Study

- Organised detailed land use survey along the railway alignment, quantified potential crop loss, assessed impacts on Sal forest, homestead plantations and wetlands, and reviewed the compensation framework.

May 1992 – February 1997 | Biologist

Bangladesh Engineering and Technological Services Ltd. (BETS) and Nature Conservation Movement (NACOM), Dhaka — North East Regional Water Management Project (NERP, FAP 6), CIDA-funded

- Assessed the status of wetland resources of north-east Bangladesh; supervised land use, ecological, limnological, wildlife and vegetation surveys; conducted ethnobiological studies on community dependence on wetland products.
- Assessed impacts of flood control, drainage and irrigation projects on wetland vegetation communities and natural resources, and recommended remedial and design measures; contributed to the Initial Environmental Examination for the North East Regional Water Management Plan and to vegetation and forestry chapters of EIAs for water resources projects.
- Prepared plantation plans and erosion protection designs, and developed and delivered training materials for a range of stakeholder groups.

July 1990 – April 1992 | Biologist / Botanist

Nature Conservation Movement (NACOM), Dhaka

- Conducted inventories and ecological characterisation of wetlands and forests, organised ecological surveys and mid-winter waterfowl census, and produced awareness and training materials.

July 1989 – June 1990 | Research Assistant

Biotechnology Research Centre, University of Dhaka

- Supported research on population genetics of aquatic plants and genetic aberration in fish resulting from water toxicity.

SELECTED PUBLICATIONS

Peer-reviewed journal articles and working papers (most relevant)

- Ahammad, R., Hossain, M. K., Sobhan, I., Hasan, R., Biswas, S. R., & Mukul, S. A. (2023). Social-ecological and institutional factors affecting forest and landscape restoration in the Chittagong Hill Tracts of Bangladesh. *Land Use Policy*, 125, 106478.

- Mukhopadhyay, A., Wheeler, D., Dasgupta, S., Dey, A., & Sobhan, I. (2019). Mangrove spatial distribution in the Indian Sundarbans: predicting salinity-induced migration in a changing climate. *Journal of Management and Sustainability*, 9(1).
- Mukhopadhyay, A., Wheeler, D., Dasgupta, S., Dey, A., & Sobhan, I. (2018). Aquatic salinization and mangrove species in a changing climate: impact in the Indian Sundarbans. *World Bank Policy Research Working Paper* 8532.
- Dasgupta, S., Huq, M., Sobhan, I., & Wheeler, D. (2018). Sea-level rise and species conservation in Bangladesh's Sundarbans region. *Journal of Management and Sustainability*, 8(1).
- Dasgupta, S., Huq, M., Mustafa, M. G., Sobhan, M. I., & Wheeler, D. (2017). The impact of aquatic salinization on fish habitats and poor communities in a changing climate: evidence from southwest coastal Bangladesh. *Ecological Economics*, 139, 128-139.
- Dasgupta, S., Sobhan, I., & Wheeler, D. (2017). The impact of climate change and aquatic salinization on mangrove species in the Bangladesh Sundarbans. *Ambio*, 46, 680-694.
- Rahman, S., Rahman, S. M., Motaleb, M. A., Sobhan, I., & Khan, N. A. (2012). Geo-spatial techniques and route and corridor mapping of Asian elephants: a participatory initiative for conservation. *Bangladesh Forestry Congress*.
- Cho, M. A., Skidmore, A. K., & Sobhan, I. (2009). Mapping beech (*Fagus sylvatica* L.) forest structure with airborne hyperspectral imagery. *International Journal of Applied Earth Observation and Geoinformation*, 11(3), 201-211.
- Cho, M. A., Skidmore, A., Corsi, F., Van Wieren, S. E., & Sobhan, I. (2007). Estimation of green grass/herb biomass from airborne hyperspectral imagery using spectral indices and partial least squares regression. *International Journal of Applied Earth Observation and Geoinformation*, 9(4), 414-424.
- Sobhan, I. (2007). Species discrimination from a hyperspectral perspective. PhD thesis, ISBN 978-90-8504-809-1.
- Sobhan, M. I., Alam, Sk. S., & Zaman, M. A. (1991). Cytogenetics of Commelinaceae XI: four taxa of *Tradescantia*. *Bangladesh Journal of Botany*, 20(2), 199-206.

Lead author — reports and management plans

- Biodiversity of Tanguar Haor, a Ramsar Site — Volume 1: Wildlife; Volume 2: Flora
- Bangladesh's Fourth National Report to the Convention on Biological Diversity
- Environmental Management Plan, Dampara Water Management Project

Contributing author

- Coping with Climate Change in the Sundarbans: Lessons from Multidisciplinary Studies (World Bank), ISBN 978-1-4648-1587-4
- Biodiversity Management Plan for the Bangladesh Sundarbans
- Adaptation Management Plan for Climate Change for four coastal districts
- Impact assessment of the proposed Indian River Linking Project for inter-basin water transfer (Component 2: Environmental Support)
- Coastal Land Use Zoning in the South-West of Bangladesh (SEMP Component 1.4.3)

- EIA, Gorai River Restoration Project; EIA/SIA, Khulna–Jessore Drainage Rehabilitation Project
- North East Regional Water Management Plan; Wetland Resources Specialist Study for North East Bangladesh

REFERENCES

Available on request.